

Retrieval Quality in Grounded Question Answering

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1. Introduction

Grounded question answering systems are judged less by the fluency of their prose than by whether the passage behind each claim genuinely supports it. A system that answers well from the wrong passage is harder to trust than one that declines to answer at all, because the failure is invisible to the reader.

We evaluate three families of retrieval configuration against a corpus of technical documentation, and report the effect of each on citation accuracy. Our interest is practical rather than theoretical: every configuration we test is one a small team could deploy without a dedicated vector database.

2. Background

Dense retrieval over document fragments has become the default architecture for question answering over private corpora. The literature has converged on a broad shape, but the parameters that matter most in practice are rarely reported.

2.1 Chunking

Chunks of roughly 512 tokens with 64 tokens of overlap outperformed both 256-token and 1024-token configurations on answer coverage. Smaller chunks fragmented multi-sentence explanations; larger chunks diluted the query signal with unrelated material from adjacent sections.

Splitting on document structure rather than on a fixed character count produced chunks with better reproducibility across re-indexing runs, which matters because the chunk identifiers are regenerated on every run.

2.2 Embeddings

All embeddings were produced at 1024 dimensions and stored without further normalization. The document and query sides of the pipeline must use the same model and the same dimension; where they diverged in a pilot run, retrieval degraded silently rather than failing, and no automated test detected the regression.

3. Methods

3.1 Corpus

The sampling frame was drawn from 4,318 publicly available support articles collected between March and August 2024. Articles shorter than two hundred words were excluded, as were duplicates identified by exact text match.

3.2 Sampling

From the sampling frame we drew a stratified random sample of six hundred articles, balanced across the eleven product areas represented in the corpus. Two annotators independently marked the passage answering each question, and disagreements were resolved by discussion.

4. Results

Stripping running headers and footers before embedding improved mean reciprocal rank from 0.61 to 0.74. The effect was largest on documents with long running titles, where the repeated text accounted for a substantial share of every short chunk.

Configuration	MRR	Hit@3
baseline	0.61	0.72
headers stripped	0.74	0.86
structure-aware	0.79	0.90

Structure-aware splitting added a further four points of mean reciprocal rank over header stripping alone. We attribute the improvement to the reduced incidence of chunks that begin or end mid-sentence, which annotators consistently rated as harder to read when shown as a citation.

5. Discussion

The configurations that helped most were also the cheapest to implement. None of them required a change of model, and none of them required more than a few hundred lines of parsing code. We take this as evidence that retrieval quality in small systems is more often lost in preprocessing than in the choice of embedding.

A limitation of this work is that all documents were born-digital. Scanned documents, which require optical character recognition before any of this applies, were excluded from the corpus entirely.

1. Annotation guidelines and the full question set are available on request.

6. Extended results

6.1 Additional comparisons

The retrieval layer was measured against a held-out set of two hundred questions. The difference was small but consistent across product areas. The annotation protocol was compared with the annotations produced in the second round. We report the mean over five runs; the variance was negligible. The evaluation harness was tuned alongside a baseline that retrieved whole documents. The effect did not survive correction for document length. The ranking model was validated using the configuration reported in the previous

section. Annotators preferred the shorter passages in every product area.

The chunking pass was contrasted with an ablation with overlap removed entirely. No configuration recovered an answer that chunking had split in half. The citation viewer was calibrated against a lexical retriever using BM25 scoring. The difference was small but consistent across product areas. The ingestion worker was measured against a held-out set of two hundred questions. We report the mean over five runs; the variance was negligible. The embedding cache was compared with the annotations produced in the second round. The effect did not survive correction for document length.

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